Consulting_Prj

Bisola

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```
#load dataset
library(readr)
consult <- read_csv("Group3Data.csv")
```

```
## Rows: 39644 Columns: 61
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (1): url
## dbl (60): timedelta, n_tokens_title, n_tokens_content, n_unique_tokens, n_no...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
View(consult)

#check data channel via tech only
table(consult$data_channel_is_tech)
```

```
##
##     0     1
## 32298  7346
```

```
#Only 7346 obs came from tech out of 39,644 users/clients
#18.5% of the sample represented


#let new dataset contain only tech data by filtering
library(dplyr)
```

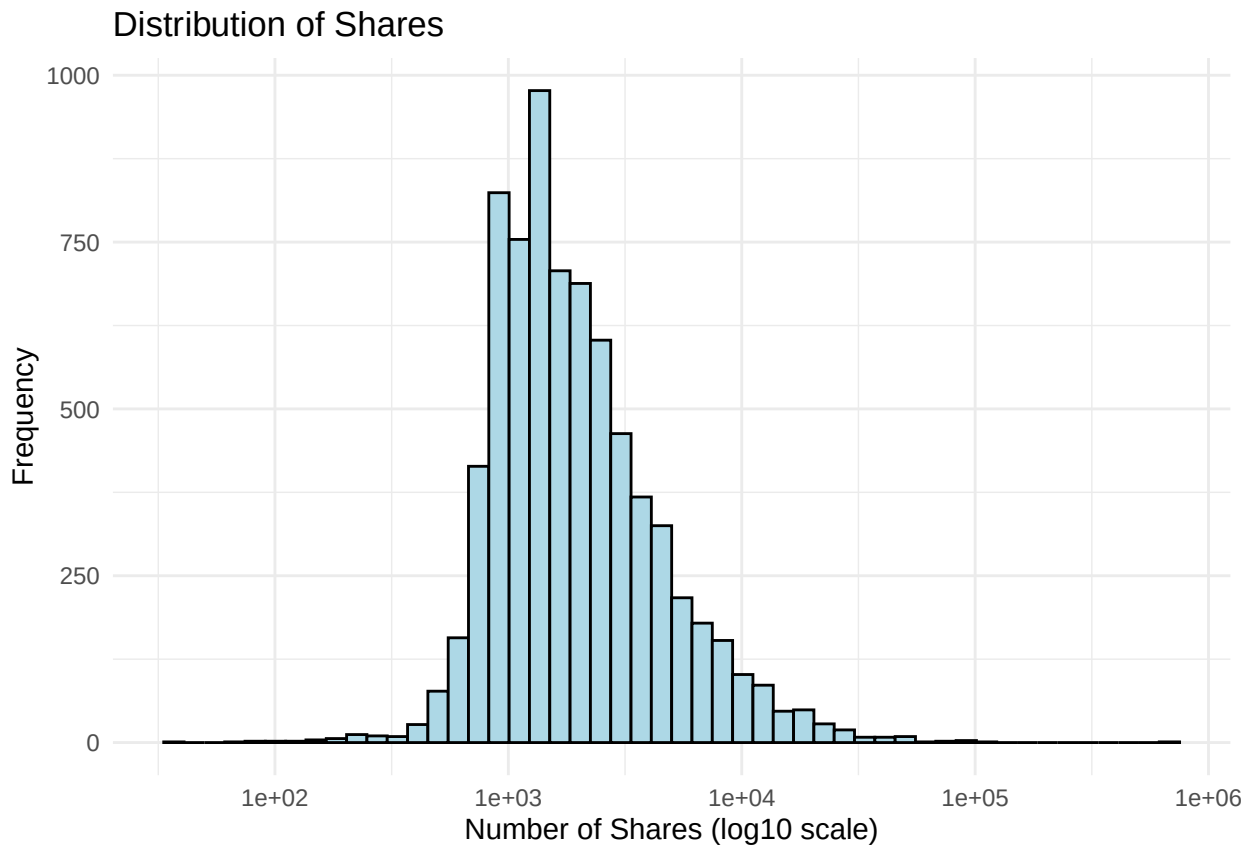
```
##
## Attaching package: 'dplyr'
##
## The following objects are masked from 'package:stats':
##
##     filter, lag
##
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
consult_tech <- consult %>%
  filter(data_channel_is_tech == 1)
```

```r
#-------------------------------------------------------------
#distribution of outcome variable (shares)
summary(consult_tech$shares)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      36    1100    1700    3072    3000  663600
```

```r
#visualization of shares
library(ggplot2)
ggplot(consult_tech, aes(x = shares)) +
  geom_histogram(bins = 50, color = "black", fill = "lightblue") +
  scale_x_log10() +
  labs(
    title = "Distribution of Shares",
    x = "Number of Shares (log10 scale)", y = "Frequency"
  ) +
  theme_minimal()
```




```r
#-------------------------------------------------------------
#combining weekdays into 1 variable
table(consult_tech$weekday_is_monday)
```

```
##
##    0    1
## 6111 1235
```

```r
library(dplyr)
consult_tech <- consult_tech %>%
  mutate(
    weekday = case_when(
      weekday_is_monday == 1 ~ "Mon",
      weekday_is_tuesday == 1 ~ "Tue",
      weekday_is_wednesday == 1 ~ "Wed",
      weekday_is_thursday == 1 ~ "Thur",
      weekday_is_friday == 1 ~ "Fri",
      weekday_is_saturday == 1 ~ "Sat",
      weekday_is_sunday == 1 ~ "Sun",
      TRUE ~ NA_character_
    )
  )

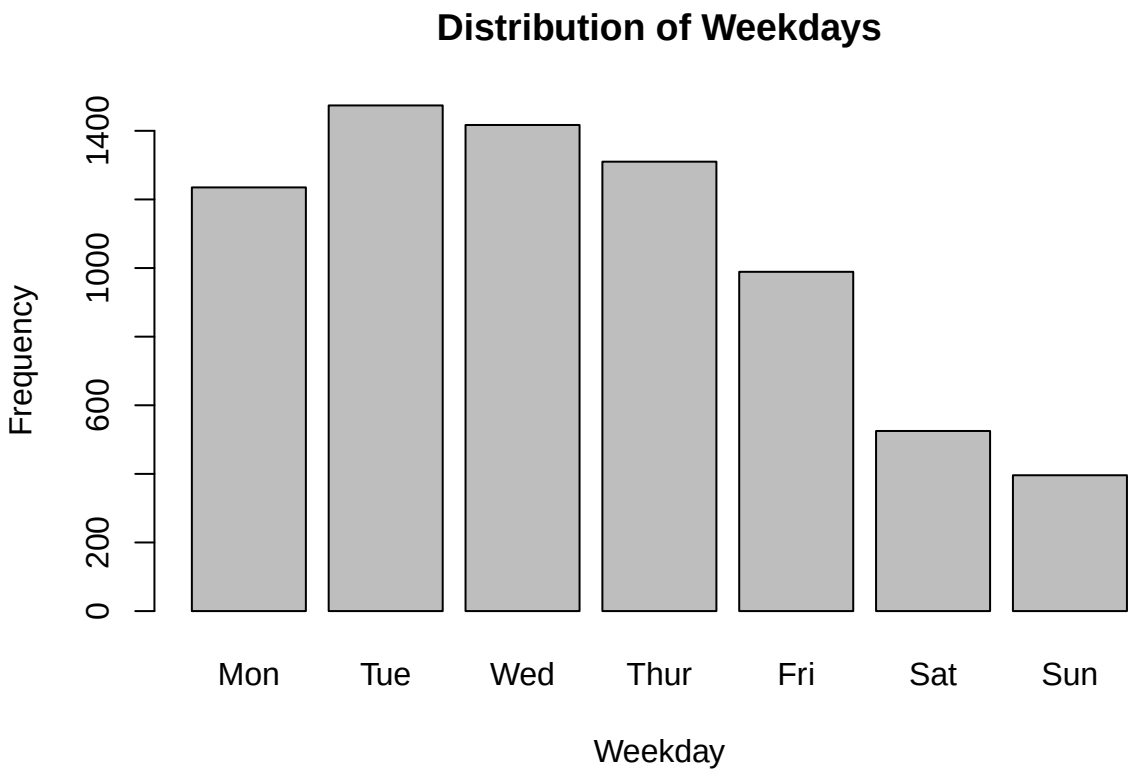

#make it a factor and order it
consult_tech$weekday <- factor(
  consult_tech$weekday,
  levels = c(
    "Mon", "Tue", "Wed", "Thur",
    "Fri", "Sat", "Sun"
  )
)



#see distribution
table(consult_tech$weekday)
```

```
##
##  Mon  Tue  Wed Thur  Fri  Sat  Sun
## 1235 1474 1417 1310  989  525  396
```

```r
#view plot
barplot(
  table(consult_tech$weekday),
  main = "Distribution of Weekdays",
  xlab = "Weekday",
  ylab = "Frequency"
)
```




```
#Polarity
summary(consult_tech$avg_negative_polarity)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -1.0000 -0.2850 -0.2220 -0.2288 -0.1667  0.0000
```

```
summary(consult_tech$avg_positive_polarity)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0000  0.3139  0.3560  0.3571  0.4013  1.0000
```

```
#Visuals
summary(consult_tech$num_imgs)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   1.000   1.000   4.435   6.000  65.000
```

```
summary(consult_tech$num_videos)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0000  0.0000  0.0000  0.4472  1.0000 73.0000
```

```r
summary(consult_tech$num_hrefs)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   5.000   7.000   9.417  11.000 120.000
```

```r
#sentiment
summary(consult_tech$global_sentiment_polarity)
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## -0.33912  0.09713  0.14379  0.14577  0.19102  0.65000
```

```r
#----------------------------------------------------------------------
#descriptive plots
#shares by day
library(dplyr)

shares_by_day <- consult_tech %>%
  group_by(weekday) %>%
  summarise(
    n_articles = n(),
    total_shares = sum(shares, na.rm = TRUE),
    mean_shares = mean(shares, na.rm = TRUE),
    median_shares = median(shares, na.rm = TRUE)
  )

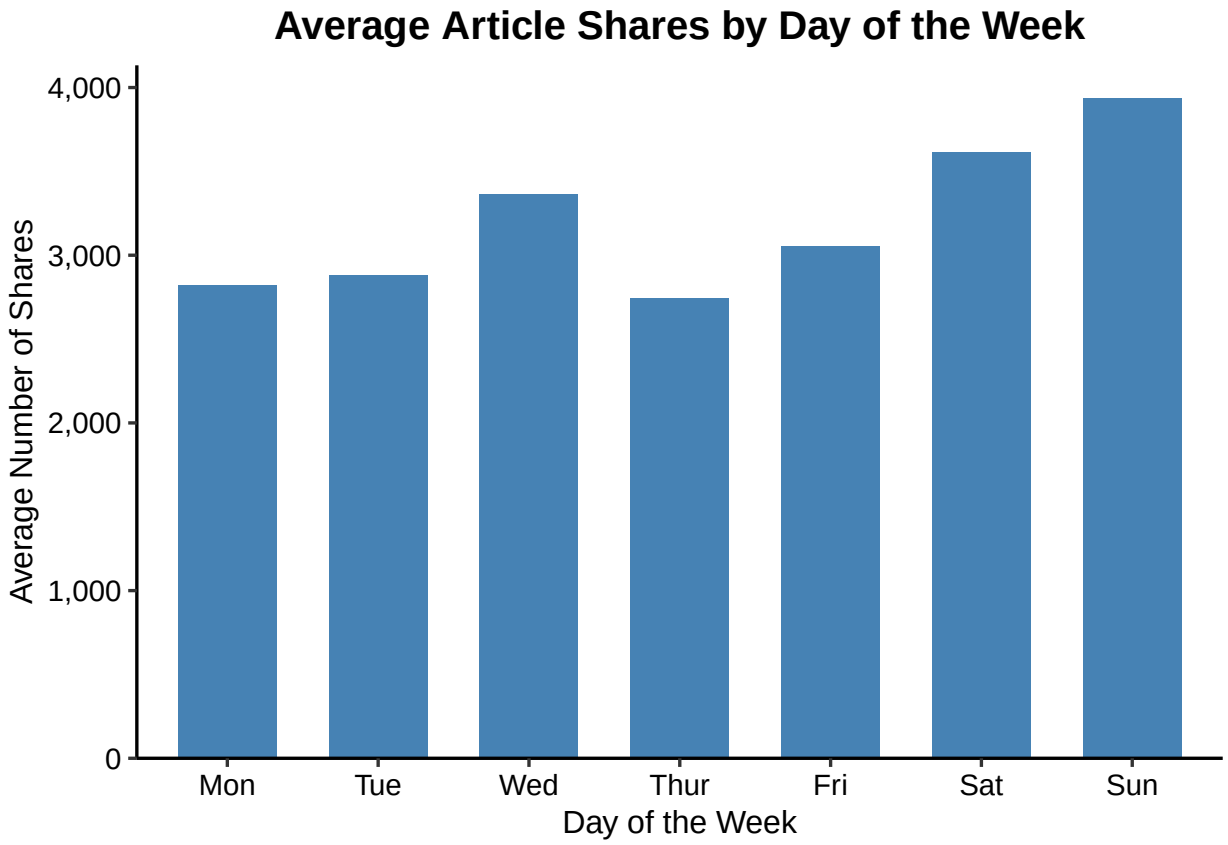
print(shares_by_day)
```

```
## # A tibble: 7 x 5
##   weekday n_articles total_shares mean_shares median_shares
##   <fct>        <int>        <dbl>       <dbl>         <dbl>
## 1 Mon           1235      3484532       2821.          1600
## 2 Tue           1474      4250146       2883.          1600
## 3 Wed           1417      4765065       3363.          1600
## 4 Thur          1310      3595351       2745.          1600
## 5 Fri            989      3017254       3051.          1800
## 6 Sat            525      1898113       3615.          2300
## 7 Sun            396      1558532       3936.          2400
```

```r
#visualization of shares by day
library(ggplot2)

ggplot(shares_by_day,
       aes(x = weekday, y = mean_shares)) +
  geom_col(fill = "steelblue", width = 0.65) +
  scale_y_continuous(
    labels = scales::comma,
    expand = expansion(mult = c(0, 0.05))
  ) +
  labs(
    title = "Average Article Shares by Day of the Week",
    x = "Day of the Week",
```

```
    y = "Average Number of Shares"
  ) +
  theme_classic(base_size = 13) +
  theme(
    plot.title = element_text(
      face = "bold",
      hjust = 0.5,
      size = 15
    ),
    axis.text.x = element_text(
      size = 11,
      color = "black"
    ),
    axis.text.y = element_text(
      size = 11,
      color = "black"
    ),
    axis.title = element_text(size = 12),
    panel.grid = element_blank()
  )
```




```
#heatmap correlation matrix between Number of links, Number of images,
#Number of videos and shares

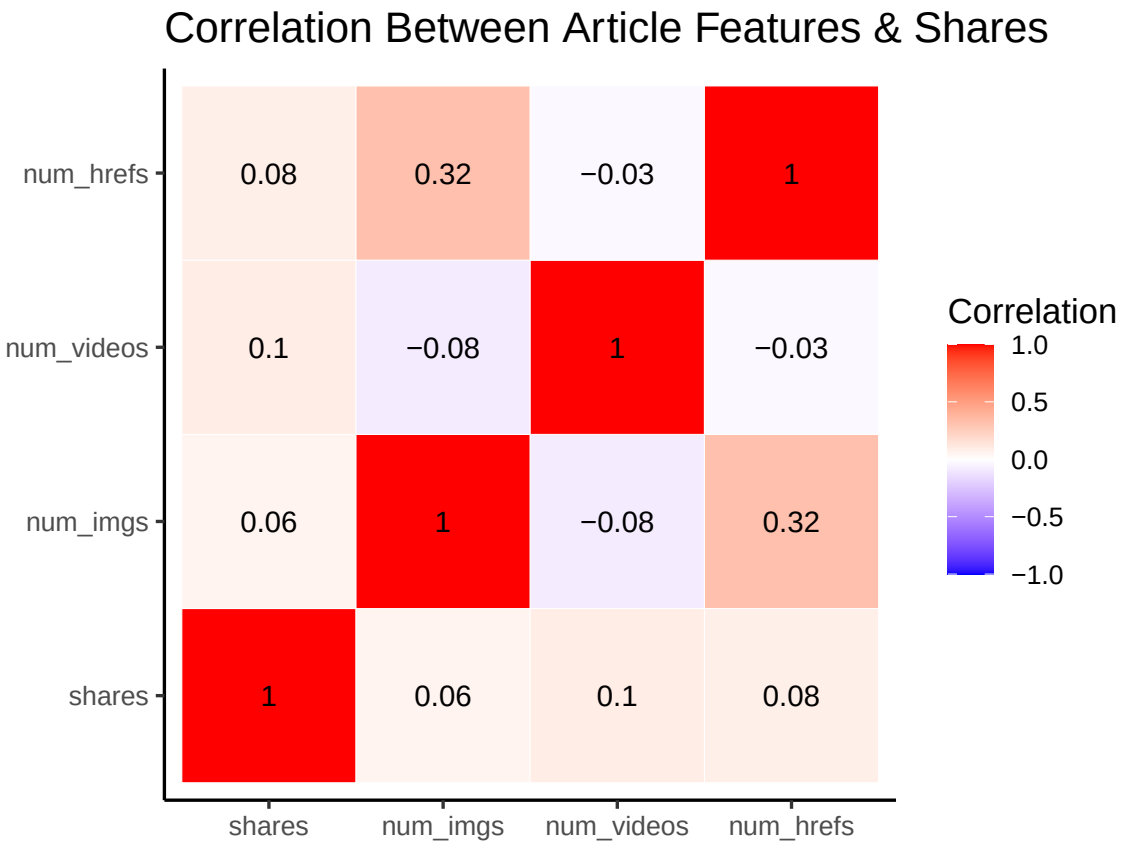
library(ggplot2)
```

```r
library(dplyr)
library(tidyr)

cor_data <- consult_tech %>%
  select(shares, num_imgs, num_videos, num_hrefs) %>%
  cor(use = "pairwise.complete.obs",
      method = "spearman")

cor_long <- as.data.frame(as.table(cor_data))

ggplot(cor_long,
       aes(x = Var1, y = Var2, fill = Freq)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(Freq, 2)), size = 4) +
  scale_fill_gradient2(
    low = "blue",
    mid = "white",
    high = "red",
    midpoint = 0,
    limits = c(-1, 1)
  ) +
  coord_equal() +
  labs(
    title = "Correlation Between Article Features & Shares", #spearman corr matrix
    x = NULL,
    y = NULL,
    fill = "Correlation"
  ) +
  theme_classic(base_size = 13)
```


Correlation Between Article Features & Shares
num_hrefs
num_videos
num_imgs
shares
0.08
0.32
−0.03
1
0.1
−0.08
1
−0.03
0.06
1
−0.08
0.32
1
0.06
0.1
0.08
shares
num_imgs
num_videos
num_hrefs
Correlation
1.0
0.5
0.0
−0.5
−1.0